NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Physiology: Transport of Acids and Bases

Bardia Askari, PhD
Department of Biomedical Sciences
baskari@nyit.edu

You decided to go on a skiing vacation to Aspen, Colorado for your winter break. You've only skied in New York/New England all and have wanted to ski "real mountains" all your life. You and a friend land in Denver and then drive to Aspen (3 hr trip). As you get closer to Aspen, you start to get a headache and feel a little lightheaded. Ibuprofen takes the edge off but everything comes back 5-fold the next day when you take the lift to the top of the mountain for your first run.

You develop a throbbing headache, become nauseous, are very short of breath and can barely focus going down the slope. Surprisingly, your friend is feeling completely fine and is loving the whole experience. You wonder on how come they're okay while you feel like your head is in a vice. They tell you that before the trip, she'd gone to her doctor and was prescribed a drug she's never heard of before and the prescription was very affordable.

You start asking yourself what's wrong with you and whether your friend's pills can help you.

Denver: 5,280 feet (1,609 meters)
Aspen: 8,000 feet (2,400 m)

High Altitude

????

Headache
Fatigue
Shortness of breath

Mystery Drug

Session Objectives

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

At the conclusion of the session, the student should be able to:

1. Define & describe the major sources of acids and bases in the body, including metabolic processes

2. Explain the fundamentals of acid/base balance in the body.

3. Define & describe the relative roles of the various segments of the tubule in the renal processing of H^+ and HCO_3^-.

4. Define & describe the major regulatory sites and mechanisms of renal processing of H^+ and HCO_3^-.

Acid-Base Balance and the Renal Handling of Acid

NEW YORK INSTITUTE OF TECHNOLOGY

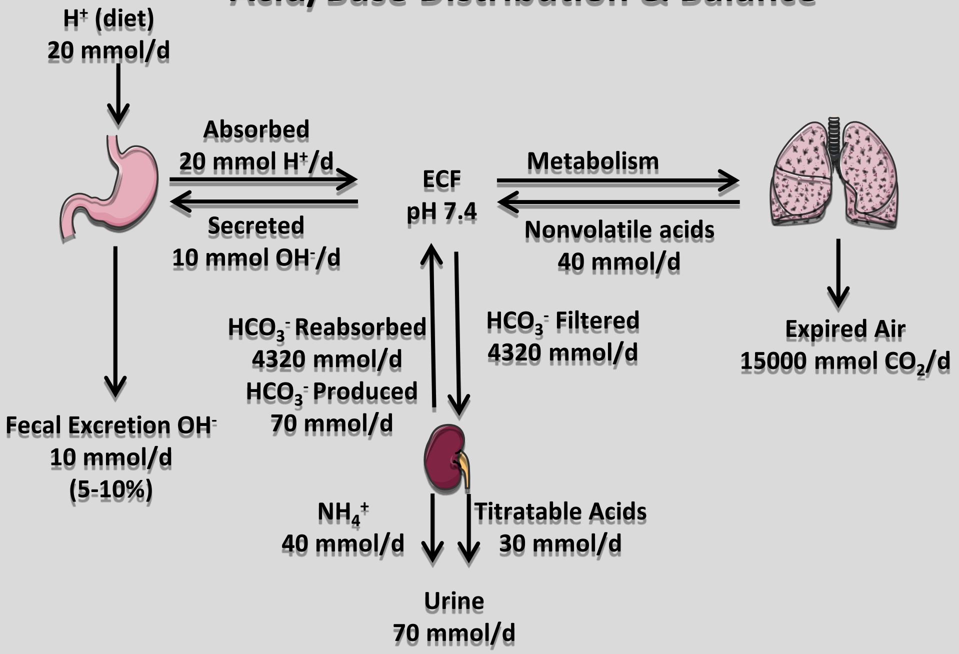
College of Osteopathic Medicine

- Acid-base imbalances are among the most important problems confronting clinicians.
- Kidneys & lungs are responsible for regulating acid-base balance.
- Via independent control of the major components of the body's buffering system: CO_2 & HCO_3^-.
- Lungs excrete the largest amount of CO_2 formed by metabolism
- Kidneys are crucial for excreting *nonvolatile* acids.
- CO_2 production: Largest source of acid
- Occurs during oxidation of carbohydrates, fats & and most amino acids
- Lungs excrete a large amount of CO_2 by diffusion across the alveolar-capillary barrier.

$$CO_2 + H_2O \underset{CA^*}{\overset{CA^*}{\rightleftarrows}} H_2CO_3 \rightleftarrows HCO_3^- + H^+$$

- Kidneys monitor the acid-base parameters of the ECF and adjust acid secretion to maintain the pH of ECF (7.37–7.42).
- Kidneys play are essential in acid-base equilibrium by neutralizing *nonvolatile* acids.
- Characteristic symptoms of renal failure:
    - Severe acidosis caused by acid retention

Acid/Base Distribution & Balance



Volatile acids: Acids derived directly from the hydration of CO_2.
Nonvolatile/Titratable acids: Acids NOT derived directly from the hydration of CO_2 (Ex: $H_2PO_4^-$).

CO_2/HCO_3^-: Major buffering system in body.
NH_3/NH_4^+: Major buffer system SYNTHESIZED in the kidney.

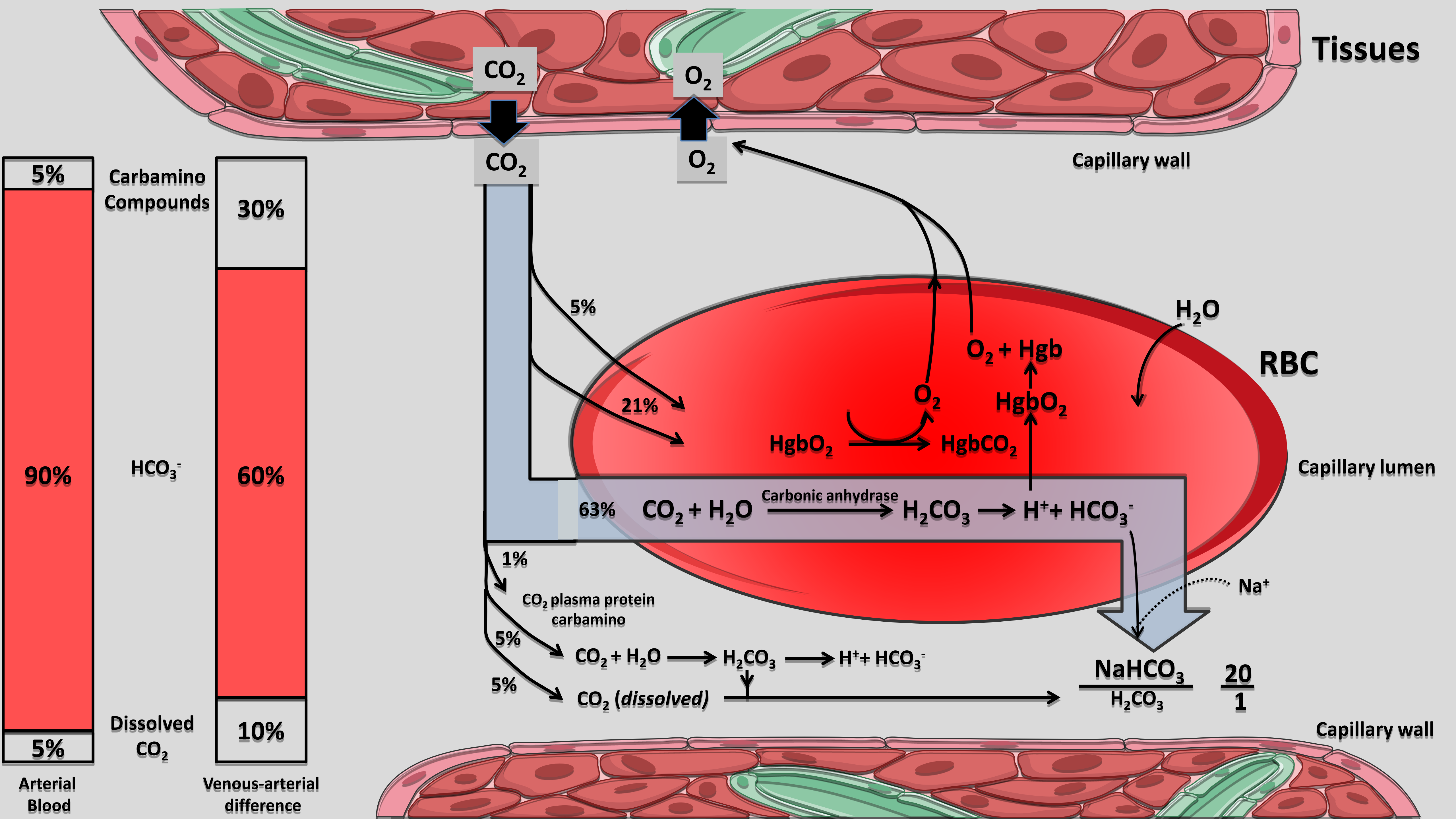
*CA: Carbonic anhydrase

Metabolic Sources of Nonvolatile Acids & Bases

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Reactions Producing CO_2	
Complete oxidation of carbohydrates & fats	$\rightarrow CO_2 + H_2O$
Complete oxidation of neutral AA	$\rightarrow$ Urea + $CO_2 + H_2O$
Reactions Producing Nonvolatile acids	
Complete oxidation of S-containing AA (Met, Cys)	$\rightarrow$ Urea + $CO_2 + H_2O + H_2SO_4 \rightarrow 2H^+ + SO_4^{-2}$
Complete oxidation of P-containing compounds	$\rightarrow H_3PO_4 \rightarrow H_2PO_4^- + H^+$
Oxidation of *cationic* amino acids (Lys, Arg)	$\rightarrow$ Urea + $CO_2 + H_2O + H^+$
Production of nonmetabolizable organic acids (uric acid, oxalic acid)	$\rightarrow HA \rightarrow H^+ + A^-$
Incomplete oxidation of carbohydrates and fats (lactic acid, ketoacids)	$\rightarrow HA \rightarrow H^+ + A^-$
Oxidation of organic anions (lactate, acetate)	$\rightarrow CO_2 + H_2O + HCO_3^-$



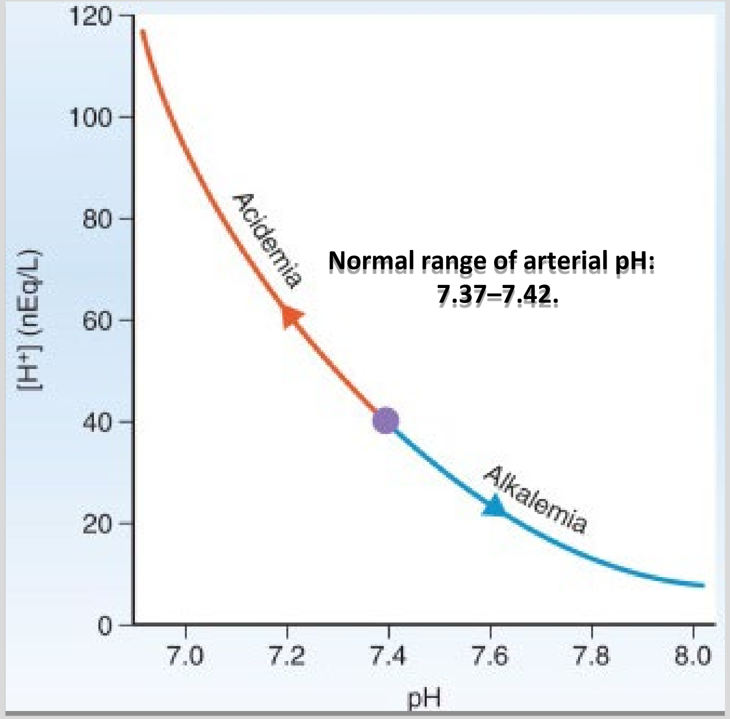
Tissues
CO2
O2
CO2
O2
Capillary wall
5%
Carbamino Compounds
30%
90%
HCO3-
60%
Dissolved CO2
5%
10%
Arterial Blood
Venous-arterial difference
5%
21%
H2O
O2 + Hgb
RBC
O2
HgbO2
HgbO2
HgbCO2
Capillary lumen
Carbonic anhydrase
63%
CO2 + H2O
H2CO3
H+ + HCO3-
1%
CO2 plasma protein carbamino
Na+
5%
CO2 + H2O
H2CO3
H+ + HCO3-
5%
CO2 (dissolved)
NaHCO3
H2CO3
20
1
Capillary wall

Kidneys Secrete Generated-Nonvolatile Acids

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

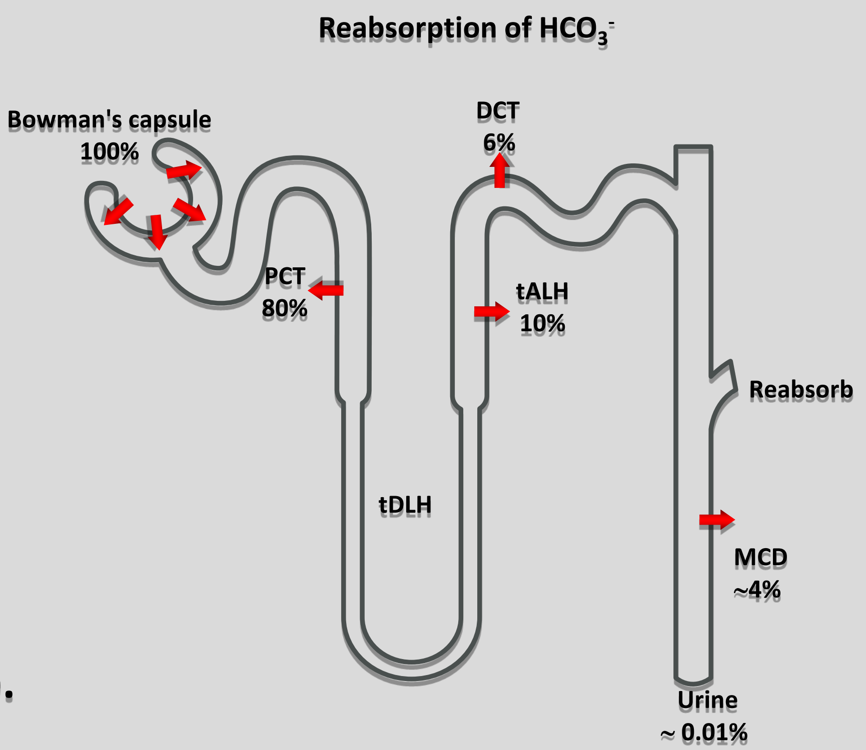
- The lungs excrete large amounts of a potential acid in the form of CO_2.
- Kidneys play an essential role in normal acid-base equilibrium.
- They are the sole effective route for neutralizing nonvolatile acid via secretion of acid into the urine.
- Prior to renal neutralization, it must retrieve ALL of filtered HCO_3^-.
- Daily filtered load of HCO_3^- = 4320 mmol.
- If this HCO_3^- was left behind in the urine $\rightarrow$ blood acid load of 4320 mmol
- Result: Catastrophic metabolic acidosis
- 70 mmol/d acid load: Produced by metabolism, diet, and intestinal losses.
- Would correspond to a urinary pH ~1.3 if not regulated.
- The lowest urinary pH is ~ 4.4
- This corresponds to an $[H^+]$ three orders of magnitude lower than required to excrete the 70 mmol/day of nonvolatile acids.
- The kidneys solve this problem by binding H^+ to:
  - Nonvolatile Acids
  - NH_3/NH_4^+ (NH_3 (renal origin) + $H^+ \rightleftarrows NH_4^+$)
  - HCO_3^-

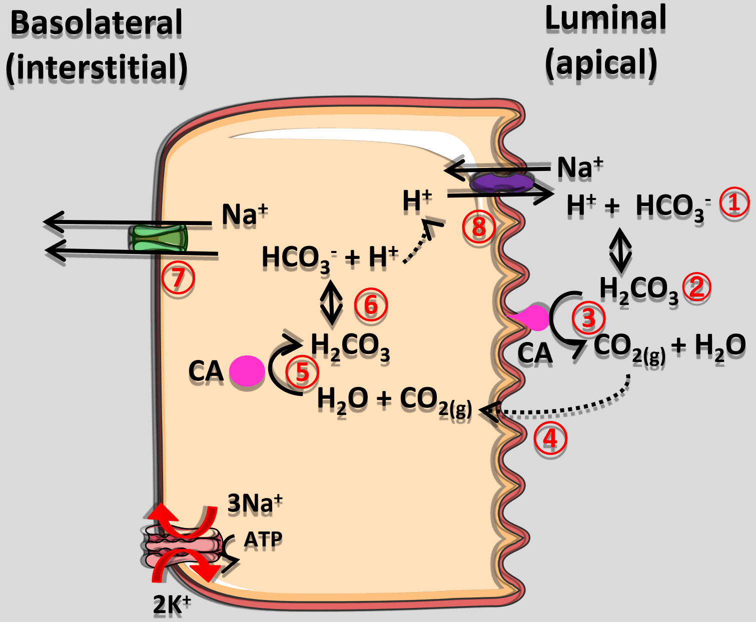


$$\text{Net } [H^+]_U \text{ excretion} = [H^+]_{Ex} \text{ bound to Nonvolatile acids} + [H^+]_{Ex} \text{ bound to } NH_3 - [HCO_3^-]_{Ex}$$

Renal Control of HCO_3^-

- HCO_3^- reabsorbed at specialized sites along the nephron.
- Basic mechanism of HCO_3^- reabsorption is the same:
  - H^+ is transported into the lumen by the tubule cell 1.
  - It is titrated w/ HCO_3^- to form H_2CO_3 (carbonic acid) 2.
  - H_2CO_3 dissociates to yield H_2O and $CO_{2(g)}$ 3.
  - Rx ($H_2CO_3 \rightarrow H_2O + CO_{2(g)}$) is too slow to convert the FL of HCO_3^-.
  - Catalyzed by carbonic anhydrase (CA).
- CA:
  - Nearly ubiquitous family of Zn-containing enzymes (at least 16 members).
  - Catalyzes the BIDERECTIONAL interconversion of CO_2 and HCO_3^-.
  - CA I: Mainly in the cytoplasm of RBC.
  - CA II: Ubiquitous cytoplasmic enzyme 5.
  - CA IV: Membrane-linked enzyme found on the outer surface of the apical membrane of the renal proximal tubule 3.
- $CO_{2(g)}$ will diffuse across the apical membrane 4 and get converted to H_2CO_3 5
- H_2CO_3 dissociates to HCO_3^- & H^+ 6.
- HCO_3^- is transported to circulation by a Na^+/HCO_3^- cotransporter 7 & H^+ is secreted into luminal space via a Na^+/antiporter 8
- A hallmark of many CAs is their inhibition by sulfonamides (e.g., acetazolamide).



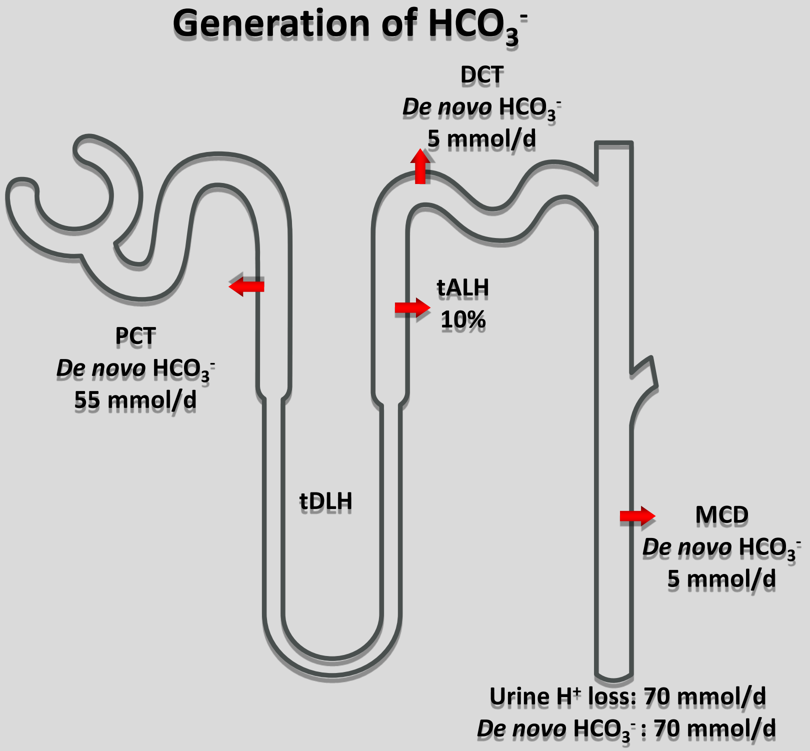


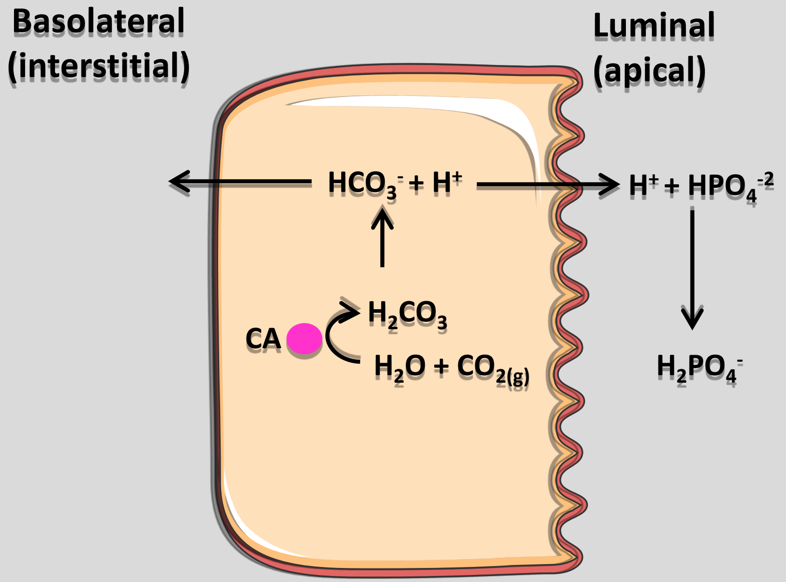
Titration of Filtered, Non-HCO_3^- Buffers

- Secreted H^+ interacts with buffers other than HCO_3^- & NH_3 (titration of non-NH_3, non-HCO_3^- buffers).
- React with HPO_4^{-2}, creatinine & urate to their conjugate weak acids (HB)

$$H^+ + B^- \rightarrow HB \text{ (titratable acid)}$$

- Kidneys excrete H^+-bound phosphate in the urine.
- ↓ pH inhibits apical Na/phosphate cotransporters (NaPi) in the PT
- ∴ transports $H_2PO_4^-$ less effectively than HPO_4^-
- Each H^+ transferred to the lumen to titrate $HPO_4^{-2} \rightarrow$ generation of new HCO_3^-
- HCO_3^- is subsequently transferred to the blood.
- Contribution to net acid excretion is dependent on 3 factors:
    - Amount of the buffer in the filtrate and urine (e.g., FL of HPO_4^{-2}).
    - pK of the buffer:
        - To be most effective at accepting H^+, buffer should have a pK that is between the pH of the glomerular filtrate and the pH of urine. (e.g., phosphate, creatinine, urate)
    - pH of the urine:
        - The lower the urinary pH, the more protonated the buffer and the greater the amount of acid excreted with this buffer.



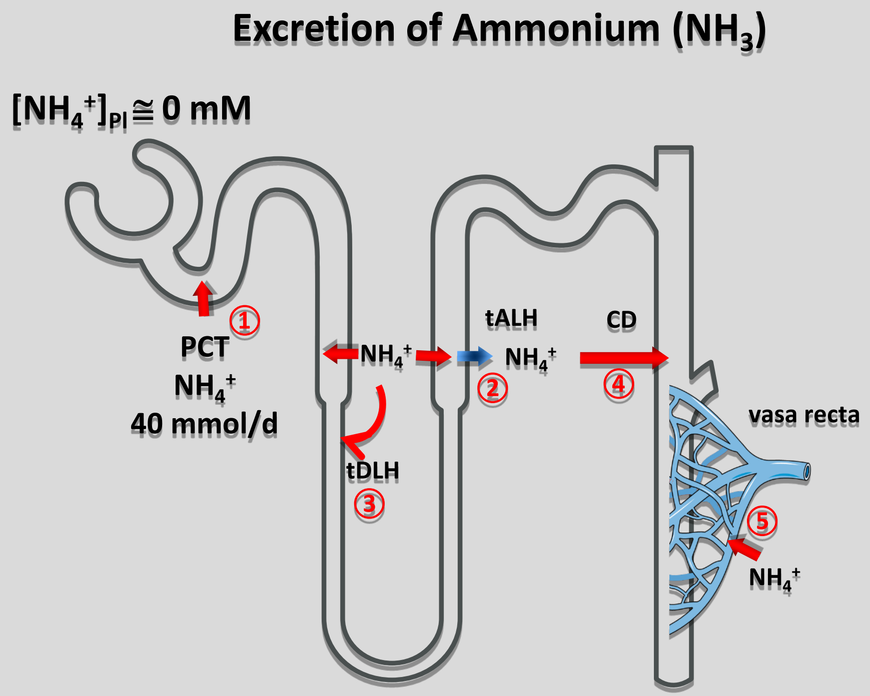


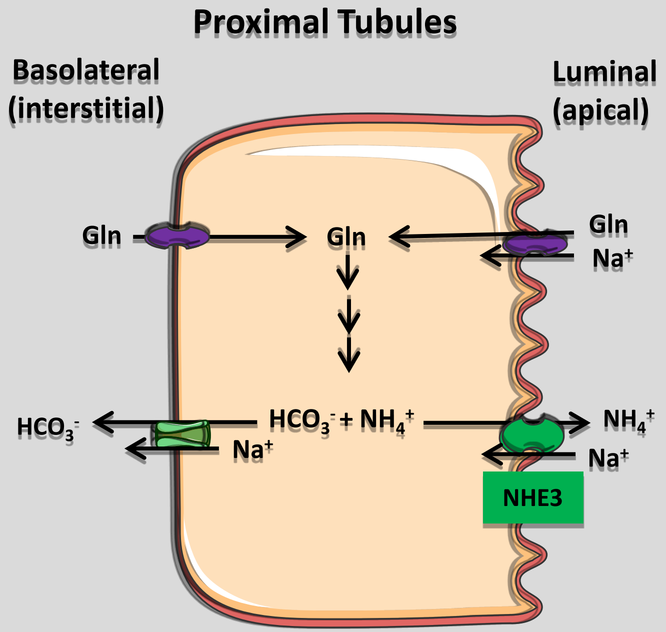
Titration of Filtered and Secreted NH_3

- NH_3: Third class of acceptors of luminal H^+.
- Unlike HCO_3^- or HPO_4^{-2}, filtration contributes only a negligible quantity of NH_3
- $[NH_3]_{Pl}$ concentration is exceedingly low.
- NH_3/NH_4^+ excretion is the result of five processes:
  - The PCT secretes ~ 40 mmol/day of NH_4^+ 1
  - tALH reabsorbs NH_4^+ & deposits it in the interstitium 2
  - Some of this interstitial NH_4^+ recycles back to the proximal tubule and thin descending limb (tDLH) 3
  - Interstitial NH_4^+ enters the lumen of the collecting duct 4
  - Interstitial NH_4^+ enters the vasa recta and leaves the kidney 5

PCT:

- Urinary NH_3 is derived from diffusion into the lumen from the PCT 1.
- Conversion of glutamine to α-ketoglutarate (α-KG) generates two NH_4^+ ions, which form two $2NH_3$ & $2H^+$.
- NH_3 enters the lumen via the apical Na^+/H^+ exchanger (NHE3).
- In addition, the metabolism of α-KG will also generate a HCO_3^- ion.
- This "new" HCO_3^- enters the blood.

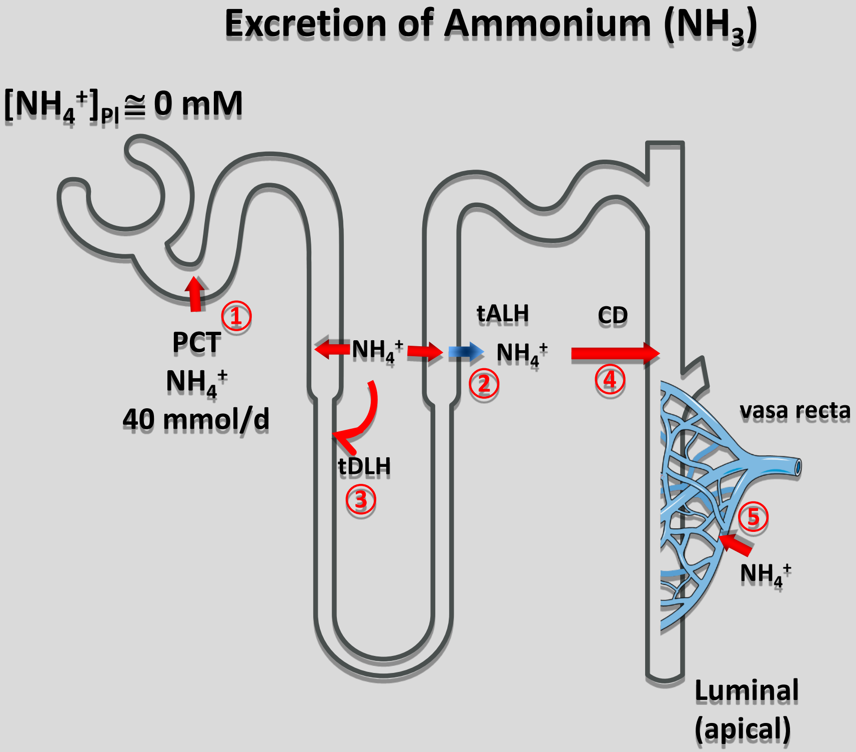


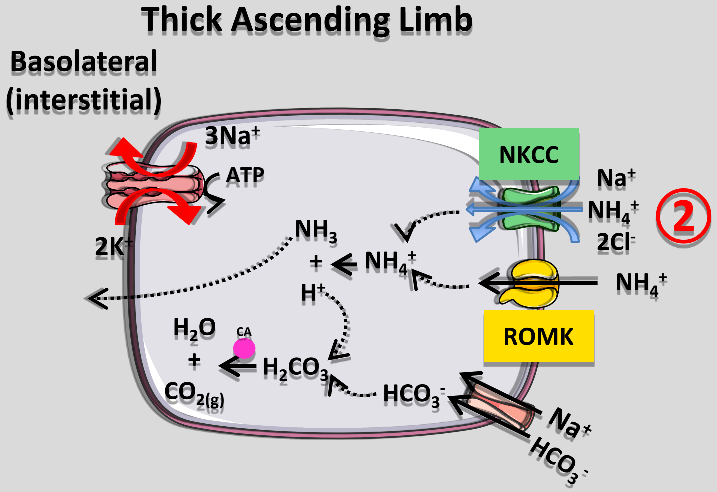


Titration of Filtered and Secreted NH_3

tALH:

- Apical membrane of the tALH has very low NH_3 permeability.
- The tALH takes up NH_4^+ across the apical membrane by using two transport mechanisms:
    - $Na^+/K^+/Cl^-$ cotransporter 2: Inhibiting cotransporter blocks a significant fraction of NH_4^+ reabsorption,
    - K^+ channels (ROMK).
- NH_3 leaves the cell across the basolateral membrane—probably as a gas → accumulation of medullary interstitial NH_4^+.
- Medullary interstitial NH_4^+ can ;
    - Dissociate into H^+ and NH_3 → enter the lumen of PST and tDLH.
    - NH_3 can also diffuse across aquaporin 1 in the basolateral & apical membranes.
    - Luminal H^+ can trap NH_3 as NH_4^+.
    - NH_4^+ can recycle between the PCT/tDLH & tALH.



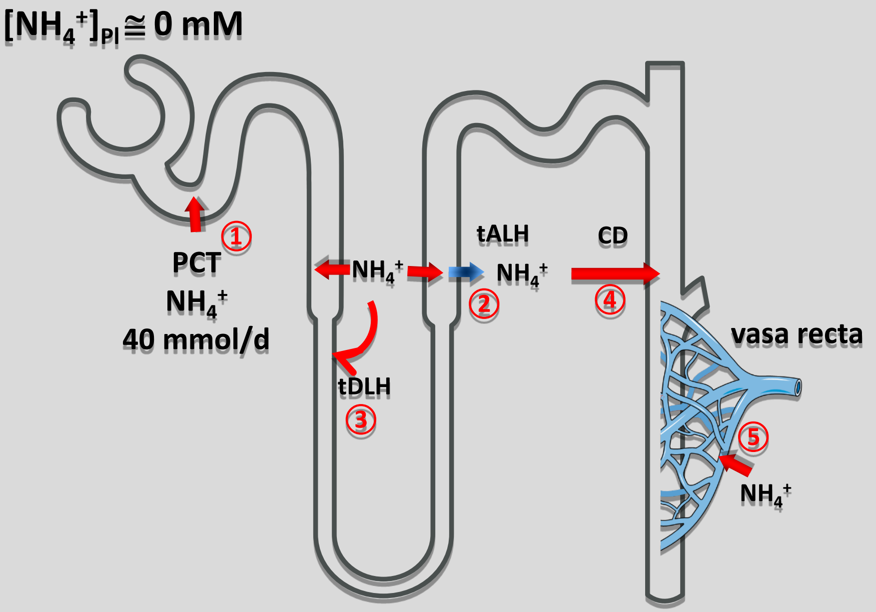


Titration of Filtered and Secreted NH_3

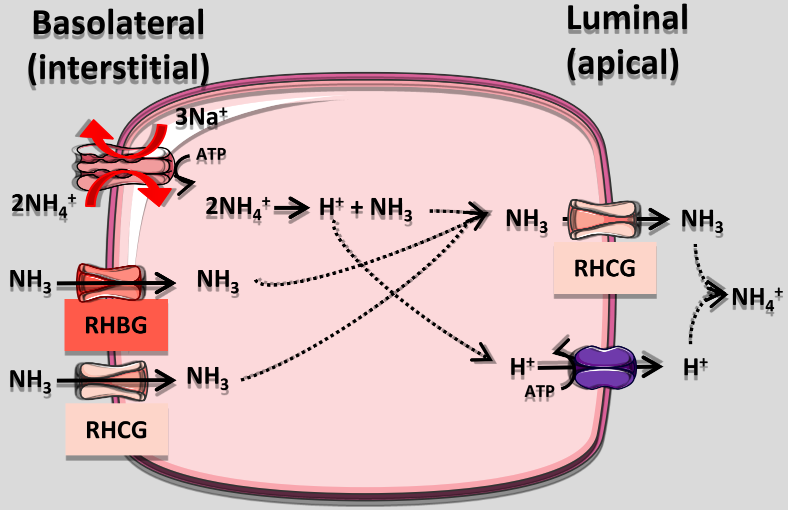
MCD:

- Some of the interstitial NH_4^+ dissociates into H^+ & NH_3 → Lumen of MCD.
- NH_3 diffuses into the cell across the basolateral membrane via gas channels (RhBG & RhCG).
- It then enters the lumen via RhCG.
- NH_3 will also combine with secreted H^+ to form NH_4^+.
- The Na^+/K^+ pump carries NH_4^+ (in place of K^+) into the MCD.
- Movement of NH_4^+ from the tALH → MCD bypasses cortical portions of the distal nephron.
- This prevents cortical portions of the distal nephron from losing NH_3 by diffusion from the lumen.
- Minimizes the entry of the toxic NH_3 back into the circulation.
- Third, a small fraction of medullary NH_4^+ enters the vasa recta (NH_4^+ washout).
- This returns the nitrogen to the systemic circulation for eventual detoxification by the liver.
- In the steady state, the buildup of NH_4^+ in the medulla leads to a sharp increase in $[NH_4^+]$ along the corticomedullary axis.

Excretion of Ammonium (NH_3)



Medullary Collecting Duct

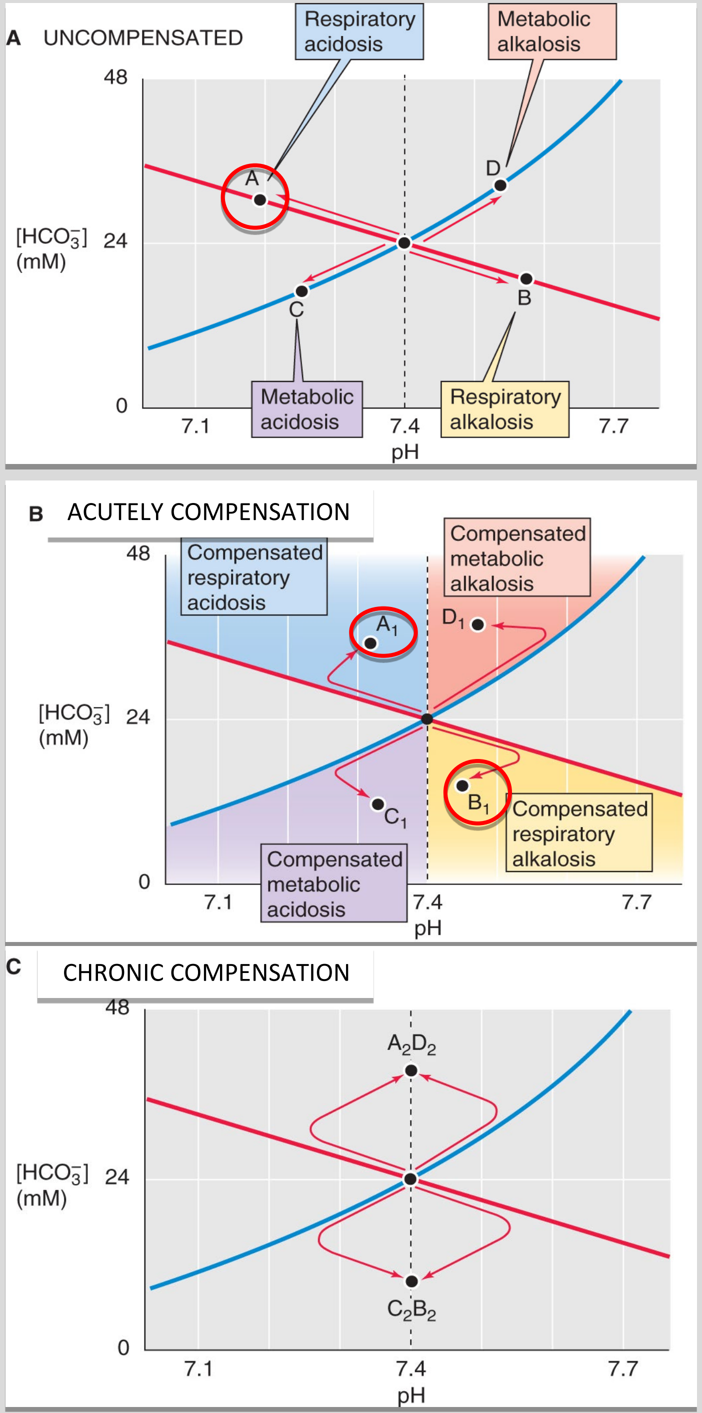


Regulation of Renal Acid Secretion

- A variety of physiological and pathophysiological stimuli modulate renal H^+/NH_3 secretion & synthesis.
- Most stimuli produce coordinated changes in acid-base transport & NH_3 production.
- 4 fundamental pH disturbances:
    - Respiratory acidosis
    - Respiratory alkalosis
    - Metabolic acidosis
    - Metabolic alkalosis
- Initial line of defense is the action of buffers (IC & EC) to minimize the magnitude of the ΔpH.
- Restoring pH to physiological values requires compensatory responses from the lungs or kidneys.

Respiratory acidosis

- When the primary disturbance is ↑arterial P_{CO2} (A)
- Compensatory response: ↑H^+ secretion → ↑ <u>production</u> of HCO_3^- via NH_3 excretion (A_1)
- The opposite occurs in respiratory alkalosis (B_1).
- Δ in H^+ secretion corrects the distorted $[HCO_3^-]/[CO_2]$ ratios



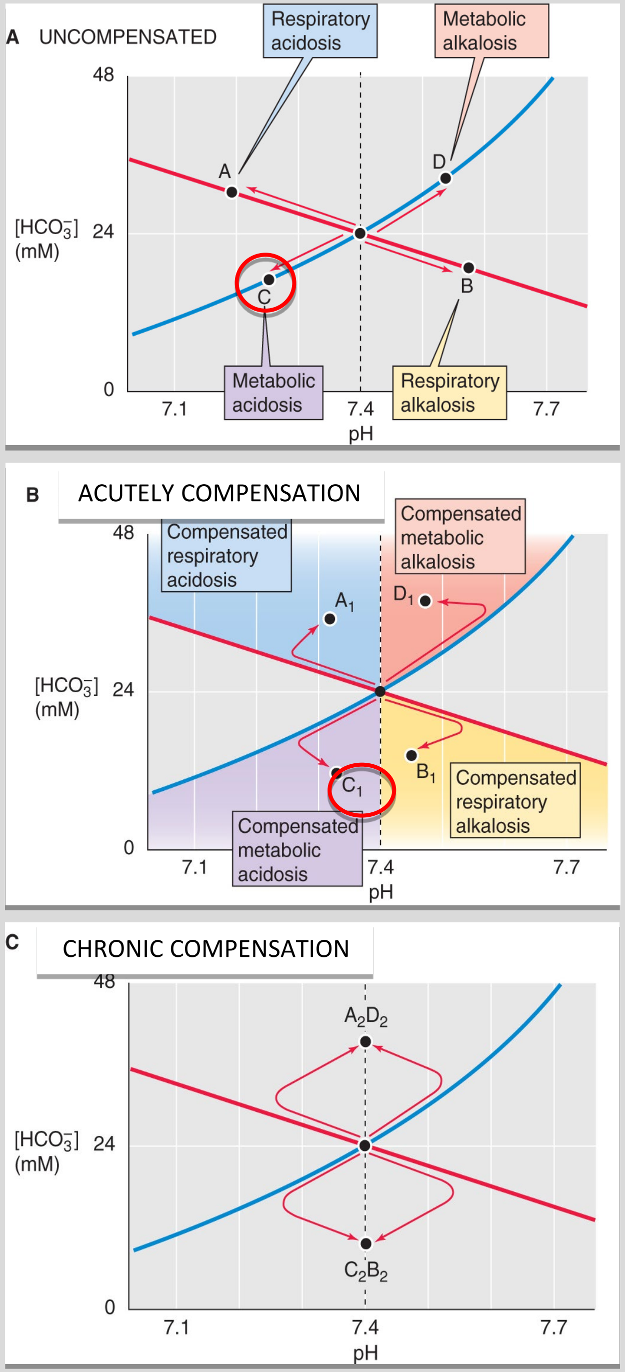
Regulation of Renal Acid Secretion

Respiratory acidosis:

- **Stimulates H^+ secretion in three ways:**
  1. **Proximal Tubules: Acute respiratory acidosis ($\uparrow P_{CO2} \rightarrow$ insertion of H^+ pumps into apical membranes $\rightarrow \uparrow H^+$ secretion**
  2. **Distal Tubule: Acute respiratory acidosis $\rightarrow$ insertion of H^+ pumps into the apical membranes $\rightarrow \uparrow H^+$ secretion.**
  3. **Chronic respiratory acidosis: Upregulation of activities of apical NHE3 and basolateral NBCe1 in proximal tubule.**

Metabolic acidosis:

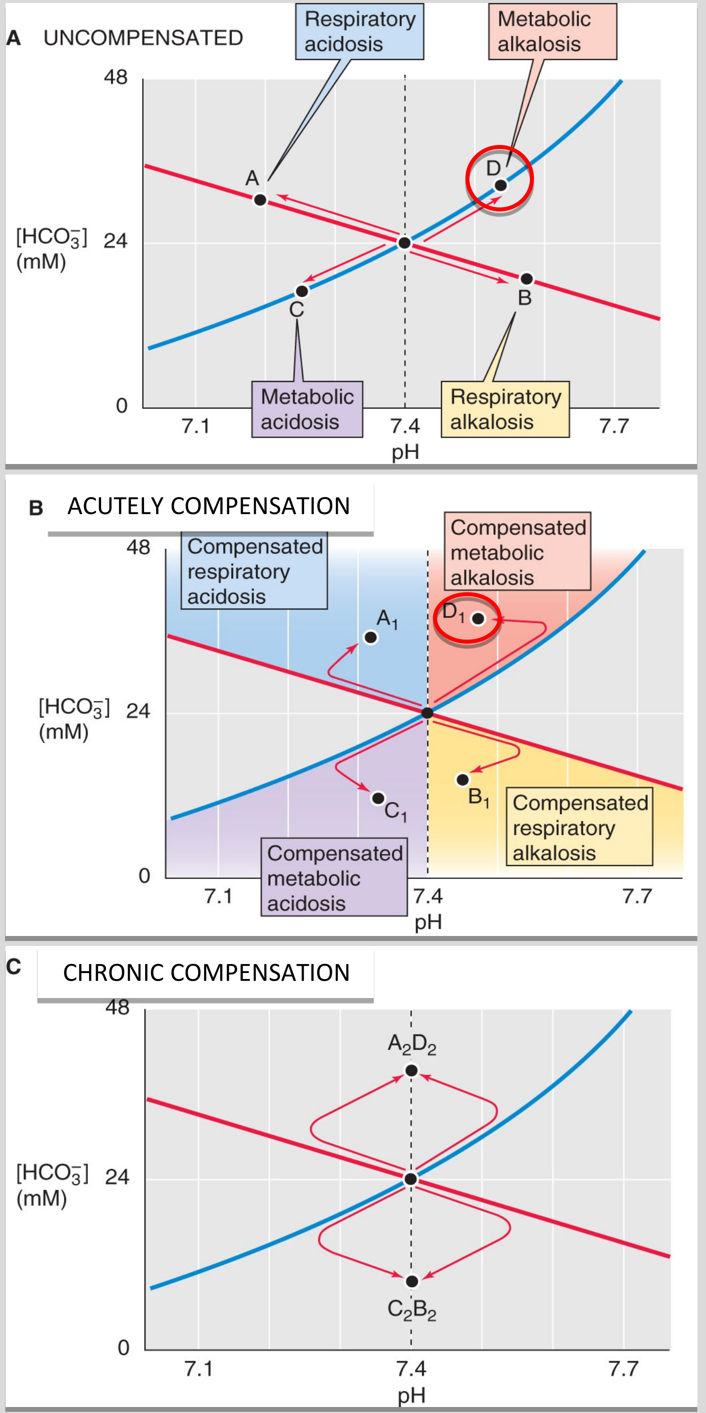
- **Acute metabolic acidosis:**
  - **Stimulates both H^+ secretion and NH_3 production**
    1. **Pulmonary compensation: Increased alveolar ventilation $\rightarrow$ CO_2 expiration $\rightarrow$ corrected $[HCO_3^-]/[CO_2]$ ratio**
    2. **Renal compensation:**
       - **Proximal Tubule: $\downarrow [HCO_3^-] \rightarrow \uparrow H^+$ secretion**
       - **Distal Tubule (Intercalated cells): $\uparrow H^+$ pump insertion into apical membranes.**
- **Chronic metabolic acidosis:**
  - **Proximal tubule: Responses are similar to chronic respiratory acidosis.**
  - **$\uparrow Na^+$-citrate uptake $\rightarrow \uparrow$Citrate metabolism $\rightarrow \uparrow HCO_3^-$ production**
  - **$\downarrow[\text{Citrate}]_{Pl} \rightarrow \downarrow[Ca^{++}\text{-Citrate}]_{EX} \rightarrow \uparrow[Ca^{++}]_U \rightarrow$ Renal calcium stones**
  - **$\uparrow$ increase NH_4^+ excretion.**



Regulation of Renal Acid Secretion

Metabolic Alkalosis

- **Proximal Tubule:**
  - **↑$[HCO_3^-]_{Pl}$ → ↓$[H^+]_{Pl}$ → ↓H^+ efflux in the PT**
- **Collecting Duct (ICT & CCT):**
  - **α-intercalated cells:**
    - **Switch from luminal $[H^+]$ secretion to $[HCO_3^-]$ secretion.**
    - **Via apical H^+ pump and basolateral Cl^-/HCO_3^- exchanger**
  - **β-intercalated cells:**
    - **Increased proportion of β IC from α IC.**
    - **They have the opposite apical/basolateral distribution of H^+ pumps and Cl^-/HCO_3^- exchangers → ↑$[HCO_3^-]_{Lumen}$**



NEW YORK INSTITUTE OF TECHNOLOGY

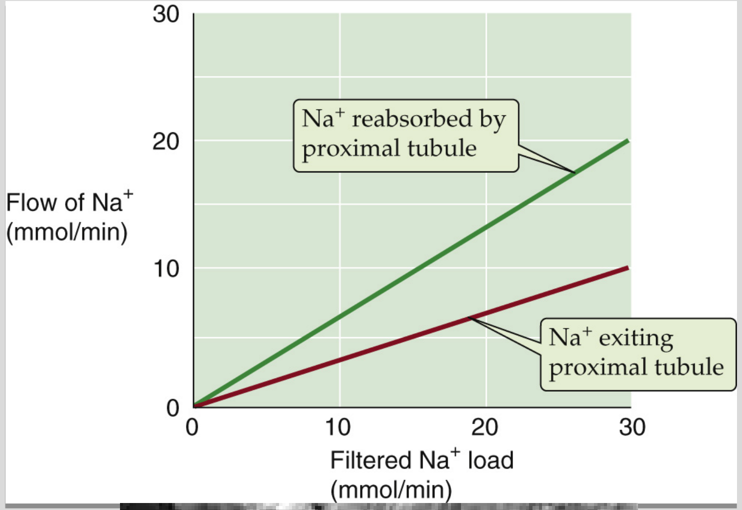
Disorder	pCO_2 (mmHg)	H^+ (mEq/L)	pH	HCO_3^- (mEq/L)	Respiratory Compensation	Renal Compensation or Correction
Normal	40	40	7.4	24	None	None
Metabolic Acidosis	↓	↑	↓	↓	Hyperventilation	↑ HCO_3^- reabsorption (correction)
Metabolic Alkalosis	↑	↓	↑	↑	Hypoventilation	↑ HCO_3^- excretion (correction)
Respiratory Acidosis	↑	↑	↓	↑	None	↑ HCO_3^- reabsorption (compensation)
Respiratory Alkalosis	↓	↓	↑	↓	None	↓ HCO_3^- reabsorption (compensation)

↓/↑ : Primary Disorder

Regulation of Renal Acid Secretion

Glomerular Filtration Rate:

- ↑ GFR increases HCO_3^- delivery to the tubules → enhances reabsorption (glomerulotubular balance).
- Similar to Na^+ balance
- ↑HCO_3^- delivery → ↑pH_{Lumen} → ↑NHE3 & H pumps in PT → enhanced H^+ secretion
- ↑Flow also increases the shear force that acts on the central cilium present on every proximal-tubule cell.
- The more the cilium bends → the greater the signal to increase the reabsorption of solutes.
- GT balance is important because it minimizes HCO_3^- loss
- ∴ ↑GFR → Development of a metabolic acidosis.
- Conversely, GT balance of HCO_3^- reabsorption also prevents metabolic alkalosis when GFR decreases.
- Flow dependence of HCO_3^- reabsorption also accounts for the stimulation of H^+ transport that occurs after uninephrectomy (i.e., surgical removal of one kidney)
- GFR in the remnant kidney rises in response to the loss of renal tissue.



Primary cilium: Flow sensor

Regulation of Renal Acid Secretion

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

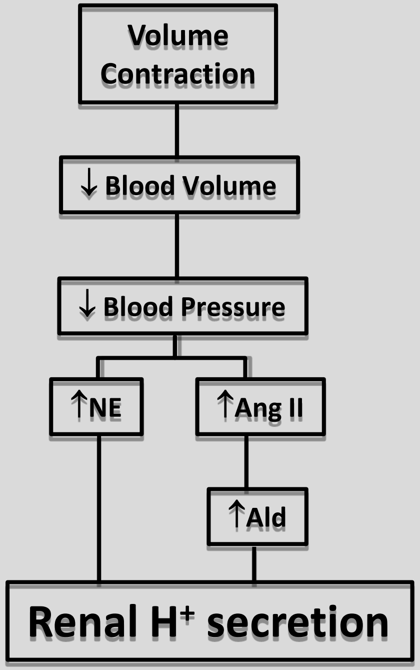
Volume Contraction:

- ↓ Extracellular volume → ↑ RAAS & SNS activity → renal H^+ secretion
- Proximal Tubule:
    - Ang II and NE stimulate Na^+/H^+ exchanger in the proximal tubule.
    - Volume contraction → Na^+ reabsorption & H^+ secretion.
- Distal Tubule:
    - AnG II stimulates acid secretion by α-intercalated cells.
- Chronic volume depletion increases aldosterone levels
    - Collecting Ducts: Aldo enhances H^+ secretion in cortical and medullary collecting ducts.
    - Associated with changes in plasma K^+

Volume Expansion:

- ↑ Extracellular volume →↓ RAAS & SNS activity → renal H^+ retention

- Thus, the regulation of effective circulating volume takes precedence over the regulation of plasma pH.



Regulation of Renal Acid Secretion

Hypokalemia:
- K^+ depletion → ↑renal H^+ secretion → metabolic alkalosis.

Proximal Tubule:
- Chronic acidification:
  - ↑Na^+/H^+ exchange (apical) & Na^+/HCO_3^- (basolateral) cotransport → ↓urinary pH
  - K^+ depletion → NH_3 synthesis & NH_4^+ excretion

Collecting Duct (ICT/CCT):
- ↑apical K^+/H^+ exchanger (αIC) → ↑H^+ secretion

Hyperkalemia
- Often associated with metabolic acidosis.

Proximal Tubule:
- ↑pH_{IC} → lower NH_4^+ synthesis → reduced NH_4^+ excretion

Thick Ascending Limb of the loop of Henle:
- $[K^+]_L$ → competition w/ NH_4^+ for uptake by NKCC & ROMK → ↓NH_4^+ reabsorption → ↓NH_4^+ in medullary interstitium → ↓NH_3 for diffusion into collecting duct

Medullary collecting duct.
- ↑$[K^+ +]$ in the medullary interstitium → competes with NH_4^+ uptake → ↓NH_4^+ excretion → Acidosis

Chronic Acidosis in Proximal Tubule

↓ pH_{EC}

⇓

↓ pH_{IC}

⇓

↑PKC ↑Tyr Kinase ↑Immediate Early Genes

⇓

↑NHE3 ↑NBCe1 ↑Na^+/Citrate Cotransporter ↑NH_3 synthesis

⇓

NH_3 synthesis & NH_4^+ excretion

Regulation of Renal Acid Secretion

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Corticosteroids:

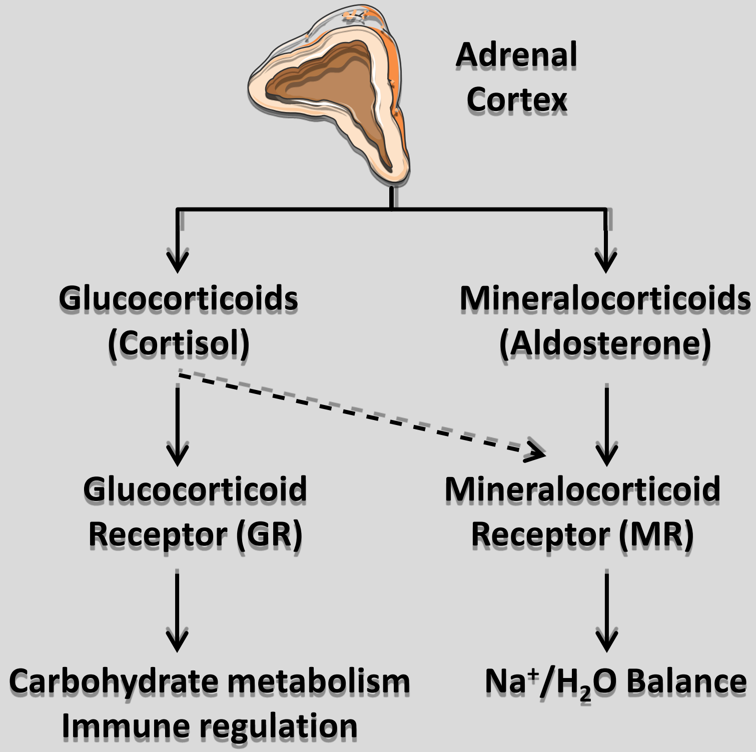
- Glucocorticoids (GC) and mineralocorticoids (MC) stimulate acid secretion.
- Chronic adrenal insufficiency → acid retention → metabolic acidosis (life-threatening)
- Both GCs & MCs H^+ secretion at different sites along the nephron.

Glucocorticoids (e.g., cortisol):

- ↑Na^+/H^+ exchanger(proximal tubule) → ↑H^+secretion
- ↓PO_4^{3-} reabsorption → ↑luminal availability of buffer anions for titration

Mineralocorticoids (e.g., aldosterone):

- ↑H^+ secretion by:
    - Directly ↑ secretion (CD) via ↑activity of the H^+ pump (apical) & basolateral Cl^-/HCO_3^- exchanger.
    - Indirectly ↑H^+ secretion by ↑Na^+ reabsorption in the CD→↑lumen-negative voltage → ↑H^+ pump in α-intercalated cells→ Acid secretion
    - Chronic elevation & accompanied by high Na^+ intake → K^+ depletion → indirectly increase H^+ secretion



Regulation of Renal Acid Secretion

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Diuretics

- The effects of diuretics on renal H^+ secretion vary among diuretics.
- Depends on site & mechanism of action.
- Diuretics fall broadly into two groups:
  - Promote excretion of alkaline urine
    - CAI and K^+-sparing diuretics
    - CAI inhibitors→ alkaline urine via ↓H^+ secretion.
    - Greatest effect is in the PT
    - ↓H^+ secretion by tALH & DCT
    - K^+-sparing diuretics (amiloride & triamterene):
    - Inhibit ENaC in CD →Lumen is more positive→ difficult for H^+ secretion into the lumen.
    - Spironolactone: ↓H^+ secretion by blocking Aldo.
  - Promote excretion of acidic urine (often inducing alkalosis)
    - Loop & Thiazide diuretics
    - They act by:
      - Volume contraction
      - Enhance Na^+ delivery to the CD→ ↑Na^+ uptake→↑ lumen-negative voltage → ↑H^+ secretion
      - K^+ wasting → Hypokalemia → ↑H^+ secretion

Nephron Site	Drug	Diuretic Class	Target
PCT	Acetazolamide	Carbonic Anhydrase Inhibitors (CAI)	Carbonic Anhydrase
tALH	Furosemide Bumetanide Ethacrynic acid	Loop diuretics	$Na^+/K^+/Cl^-$ cotransporter (NKCC2)
DCT	Metolazone	Thiazides	Na^+/Cl^- cotransporter (NCC)
CCT	Amiloride Triamterene	K^+-Sparing	Na^+ channel (ENaC)
CCT	Spironolactone Eplerenone	K^+-Sparing	Mineralocorticoid receptor (MR)
H_2O permeable segments	Mannitol	Osmotic diuretics	

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

High Altitude → ↓O_2 pressure (PO_2) ↓$PaCO_2$ → ↑pH_{Blood} → ↑HCO_3^- in Urine

Mystery

Drug ???? How & Why??

Lecture Feedback Form

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R